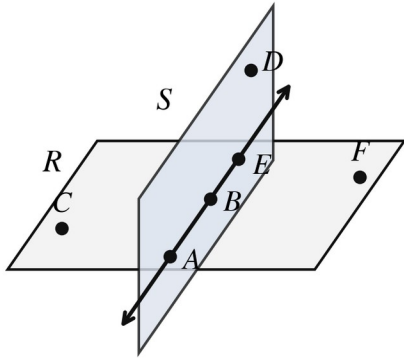


• **WORKED KEY**

Basics of Geometry • Sections 1.1 – 1.6

Every skill on the real test is here — the questions are just asked a different way. Show all your work. Circle your final answer.

Use the diagram for question 1.



1. Planes R and S intersect.
 - a) Name the intersection of the two planes.
 - b) Name three collinear points.
 - c) Name a point that is NOT coplanar with A, B, and C.

Answer: line AB (or line AE, line BE)

Answer: A, B, and E

Answer: D — it lies in plane S but not in plane R (A, B, C are all in R)

2. Sketch and label: line m intersecting plane W at exactly one point, T.
3. True or false? Any three points are collinear. Explain, and sketch a counterexample if it is false.

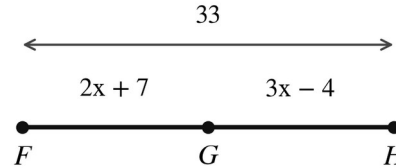
Answer: a plane (parallelogram) labeled W, a line passing through it, one marked point T where they meet; the line should poke out on both sides

Answer: False — three points can form a triangle. Three points are always COPLANAR, but only sometimes collinear.

Full Name: _____

Period: _____

Use the figure for question 4.



4. Find x. Then find FG and GH.

Answer: $(2x + 7) + (3x - 4) = 33 \rightarrow 5x + 3 = 33 \rightarrow x = 6$; $FG = 19$, $GH = 14$

5. A straight trail runs 6.8 miles from the parking lot to the summit. A lookout is on the trail, 2.9 miles from the parking lot.
 - a) How far is the lookout from the summit?
 - b) A ranger station sits on the trail at the midpoint between the lookout and the summit. How far is it from the parking lot?

Answer: $6.8 - 2.9 = 3.9$ miles

Answer: half of 3.9 is 1.95; $2.9 + 1.95 = 4.85$ miles

6. M(-1, 3) is the midpoint of $\overline{AB}$, and A has coordinates (-6, -1). Find B. Then find the length AB.
7. Is $\triangle PQR$ isosceles? P(-2, 1), Q(4, 1), R(1, -3). Use the Distance Formula and explain.

Answer: B: $x = 2(-1) - (-6) = 4$, $y = 2(3) - (-1) = 7 \rightarrow B(4, 7)$; $AB = \sqrt{(10^2 + 8^2)} = \sqrt{164} = 2\sqrt{41} \approx 12.8$

Answer: $PQ = 6$; $PR = \sqrt{(3^2 + 4^2)} = 5$; $QR = \sqrt{(3^2 + 4^2)} = 5$. Yes — $PR = QR$.

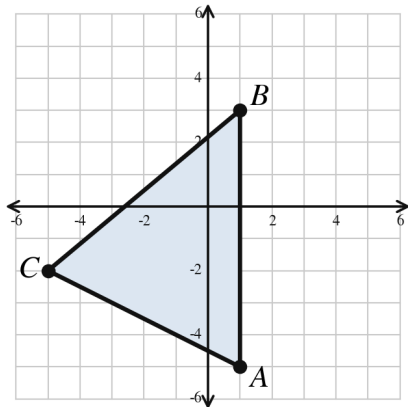
• **WORKED KEY**

Page 2

8. Sketch a concave hexagon. Then name a polygon with 10 sides.

Answer: any 6-sided figure with a “dent” (a side’s line cuts through the interior); decagon

9. Use $\triangle ABC$ below. Each square is 1 unit.



- a. Find the perimeter of $\triangle ABC$. Round to the nearest tenth.

Answer: $AB = 8$ (vertical); $AC = \sqrt{6^2 + 3^2} = \sqrt{45} \approx 6.7$; $BC = \sqrt{6^2 + 5^2} = \sqrt{61} \approx 7.8$; $P \approx 8 + 6.7 + 7.8 = 22.5$

- b. Find the area of $\triangle ABC$. (Which side is the base? Where is the height?)

Answer: base $AB = 8$ is vertical, so the height is horizontal: 6 (from $x = 1$ to $x = -5$). $A = \frac{1}{2}(8)(6) = 24$

10. Without a protractor, describe how to construct a 22.5° angle starting from a straight angle. How do you KNOW it is exactly 22.5° ?

Answer: Bisect the straight angle (90°), bisect that (45°), bisect again (22.5°). Each bisection is exact: $180 \div 2 \div 2 \div 2 = 22.5$.

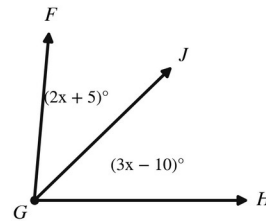
11. $\angle A$ and $\angle B$ are complementary. $m\angle A = (5x - 3)^\circ$ and $m\angle B = (2x + 9)^\circ$. Find both measures.

Answer: $7x + 6 = 90 \rightarrow x = 12$; $m\angle A = 57^\circ$, $m\angle B = 33^\circ$

Full Name: _____

Period: _____

Use the figure for question 12.



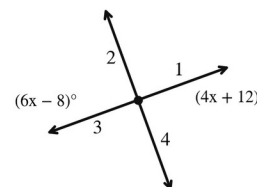
12. $m\angle FGH = 85^\circ$. Find x , then find $m\angle FGJ$ and $m\angle JGH$.

Answer: $(2x + 5) + (3x - 10) = 85 \rightarrow 5x - 5 = 85 \rightarrow x = 18$; $m\angle FGJ = 41^\circ$, $m\angle JGH = 44^\circ$

13. Ray QS bisects $\angle PQR$. $m\angle PQS = (3x + 9)^\circ$ and $m\angle PQR = (8x - 6)^\circ$. Find $m\angle PQR$.

Answer: half = whole $\div 2$: $2(3x + 9) = 8x - 6 \rightarrow 6x + 18 = 8x - 6 \rightarrow x = 12$; $m\angle PQS = 45^\circ$, so $m\angle PQR = 90^\circ$

Use the figure for question 14.



14. $m\angle 2 = (6x - 8)^\circ$ and $m\angle 4 = (4x + 12)^\circ$.

- a) Name a pair of vertical angles and a linear pair.

Answer: vertical: $\angle 2$ and $\angle 4$ (or $\angle 1$ and $\angle 3$); linear pair: $\angle 1$ and $\angle 2$ (any adjacent pair)

- b) Find x . c) Find $m\angle 1$.

Answer: $6x - 8 = 4x + 12 \rightarrow x = 10$; $m\angle 4 = 52^\circ$; $\angle 1$ and $\angle 4$ are a linear pair $\rightarrow m\angle 1 = 180 - 52 = 128^\circ$

15. $\angle C$ and $\angle D$ are supplementary. $m\angle C = (7x + 2)^\circ$ and $m\angle D = (3x - 12)^\circ$.

- a) Find both measures.

Answer: $10x - 10 = 180 \rightarrow x = 19$; $m\angle C = 135^\circ$, $m\angle D = 45^\circ$

- b) Must $\angle C$ and $\angle D$ form a linear pair? Explain.

Answer: No — supplementary only means the measures add to 180° . A linear pair must also be ADJACENT.